

pharmaniaga®

Lidocaine

Hydrochloride

2% w/v Injection



COMPOSITION

Each 10mL contains:
Lidocaine Hydrochloride 200mg
Methylparaben 10mg

DESCRIPTION

A colourless solution.

PHARMACODYNAMICS

Pharmacotherapeutic group: Local anaesthetics, ATC code: N01B B02.

Lidocaine is an amide group local anaesthetic. It inhibits the transport of sodium ions through the nerve membrane, reversibly blocking impulse transmission in nerve fibres. On excitatory membranes in the brain and myocardium, similar effects are observed. Upon absorption, lidocaine may cause central nervous system stimulation followed by depression. It affects predominantly on the myocardium where it may produce decreases in electrical excitability, conduction rate, and force of contraction. Lidocaine has a rapid onset of effect, a high rate of anaesthesia, and a low toxicity. Lower lidocaine concentrations have a lesser effect on motor nerve fibres.

PHARMACOKINETICS

The rate of absorption is dependent on dose, route of administration and perfusion at the injection site. Intercostal blockades produce the highest plasma concentrations (approx. 1.5 µg/mL per 100 mg injected), while subcutaneous injections in the abdomen produce the lowest plasma concentrations (approx. 0.5 µg/mL per 100 mg injected).The volume of distribution in steady state is 91L and plasma protein binding, which is predominantly to alpha-1-acid glycoprotein, is 65 %.

Lidocaine is eliminated predominantly through metabolism. Dealkylation to monoethyl-glycinexylidide (MEGX) is mediated mainly by cytochrome P450 3A4. MEGX is metabolised to 2,6-xylidine and glycinexylidide (GX)-2,6- dimethylaniline is metabolised further by CYP2A6 to 4-hydroxy-2,6-dimethylaniline, which is the major metabolite in the urine (80 %) and is excreted as conjugate. MEGX has a convulsant activity equivalent to that of lidocaine, while GX is devoid of convulsant activity. MEGX appears to occur in similar plasma concentrations as the parent substance. The elimination rates of lidocaine and MEGX after an intravenous bolus dose are approx. 1.5-2 and 2.5 hours respectively.

On account of the rapid hepatic metabolism, the kinetics are sensitive to all alterations in liver function. The half-life can be more than doubled in patients with impaired liver function. Impaired renal function does not affect the kinetics, but can increase the accumulation of metabolites.

Lidocaine crosses the blood-brain and placental barriers. The concentration of free lidocaine is the same in the mother and the foetus. However, the total plasma concentration is lower in the foetus, which has a lower degree of protein binding.

INDICATION

Lidocaine can be used for most major nerve blocks, e.g. sciatic and retrobulbar. Lidocaine can be used for local or regional anaesthesia and nerve block.

RECOMMENDED DOSAGE

Adults and children above 12 yrs of age

Lidocaine should only be used by physicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used.

This solution is not intended for use intravenously. Solutions of lidocaine which contain preservatives should not be used for spinal, epidural, caudal or intravenous regional anaesthesia.

Maximum recommended doses:

Lidocaine hydrochloride 20 mg/ml, 20 ml (400 mg lidocaine hydrochloride)

Table 1 Dosage Recommendations, is a guide to dosage for the more commonly used techniques in the average adult. The clinician's experience and knowledge of the patient's physical status are of importance in calculating the required dose. When prolonged blocks are used, e.g., by repeated administration, the risks of reaching a toxic plasma concentration or inducing a local neural injury must be considered.

Table 1. Dosage Recommendations					
Type of block	Conc. mg/ml	Dose		Onset (min)	Duration (h)
Major nerve block		ml	mg		
Retrobulbar	20	4	80	3-5	1.5-2
Sciatic	20	15-20	300-400	15-30	2-3

The doses given in the table are those considered to be necessary to produce successful blocks and should be regarded as a guide for use in adults. Individual variations in onset and duration occur. The figures reflect the expected average dose range needed. Standard textbooks should be consulted for factors affecting specific block techniques and for individual patient requirements.

Unnecessarily high doses of local anaesthetics are to be avoided. In general, surgical anaesthesia requires the use of the higher concentrations and doses. When blocking smaller nerves, or when a less intense block is required, the use of a lower concentration is indicated. The volume of drug used will affect the extent and spread of anaesthesia.

In order to avoid intravascular injection, aspiration should be repeated prior to and during administration of the main dose, which should be injected slowly or in incremental doses, at a rate of 100-200 mg/min, while closely observing the patient's vital functions and maintaining verbal contact. An inadvertent intravascular injection may be recognised by a temporary increase in heart rate and an accidental intrathecal injection by signs of a spinal block. If toxic symptoms occur, the injection should be stopped immediately.

The doses should be reduced for children and patients in reduced general condition.

Usual Paediatric Dose:

A reduced dosage based on body weight or surface area should be used. The dosage should be calculated for each patient individually and modified in accordance with the physician's experience and knowledge of the patient. Early signs of local anaesthetic toxicity may be difficult to detect in cases where the block is given. In order to minimise the possibility of toxic effects, the use of Lidocaine Injection 1% solution is recommended for most anaesthetic procedures involving paediatric patients. The dose should not exceed 3 mg/kg.

ROUTE OF ADMINISTRATION

Parenteral
(For intramuscular or subcutaneous injection)

This solution is not intended for use intravenously. This solution contains preservatives.

CONTRAINDICATIONS

Hypersensitivity to local anaesthetics of the amide type or to other components of the product. Lidocaine solution for injection with preservatives should not be used in patients who are hypersensitive to methyl and/or propyl parahydroxybenzoate (methyl-/propyl paraben), or to their

metabolite para amino benzoic acid (PABA). Lidocaine solution for injection with preservatives should be avoided in patients allergic to ester local anaesthetics that also has the metabolite PABA.

WARNINGS AND PRECAUTIONS

Regional or local anaesthetic procedures, except those of the most trivial nature, should always be carried out in the proximity of resuscitation equipment. Before major blocks, an intravenous cannula should be inserted before the local anaesthetic is injected. Like all local anaesthetic agents, lidocaine can cause acute central nervous and cardiovascular toxic effects in cases of use that leads to high concentrations in the blood. This applies particularly after inadvertent intravascular administration.

Caution should be exercised in patients of the following categories:

- Elderly, and patients with generally reduced general condition
- Patients with degree II or III AV block since local anaesthetics can lower the conduction capacity of the myocardium
- Patients with severe hepatic disease or severely impaired renal function with a creatinine clearance of less than 10mL/minute.
- Patients treated with class III anti-arrhythmic drugs (e.g. amiodarone) should be closely observed and ECG monitoring should be considered, since the cardiac effects of lidocaine and class III anti-arrhythmic drugs can be additive.
- Patients with epilepsy, congestive cardiac failure, bradycardia, severe shock, impaired respiratory function.

Some regional anaesthesia techniques may be associated with severe adverse reactions, as indicated below:

- Retrobulbar injections can in rare cases reach the cranial subarachnoid space and cause e.g. temporary blindness, cardiovascular collapse, apnoea and convulsions. These symptoms must be treated immediately.

- Retro and peribulbar injections with local anaesthetics involve some risk of persistent ocular muscle dysfunction. The primary causes include trauma and/or local toxic effects on muscles and/or nerves. The severity of such tissue reactions is related to the degree of trauma, the concentration of the local anaesthetic, and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used. Vasoconstrictors may aggravate tissue reactions and should be used only when indicated.

- There have been literature post-marketing reports of chondrolysis in patients receiving postoperative intra-articular continuous infusion of local anaesthetics. The majority of reported cases of chondrolysis have involved the shoulder joint. Due to multiple contributing factors and inconsistency in the scientific literature regarding the mechanism of action, causality has not been established. Intra-articular continuous infusion is not an approved indication for lidocaine.

The prime causes are traumatic nerve injury and/or local toxic effect on muscles and nerves by injected local anaesthetic. The extent of this tissue damage depends on the degree of trauma, the concentration of the local anaesthetic, and the duration of exposure to the local anaesthetic. For this reason, the lowest effective dose should be chosen.

- Inadvertent intravascular injection in the head and neck regions can cause cerebral symptoms even with low doses.

Lidocaine solution for injection is probably porphyrinogenic and should only be prescribed to patients with acute porphyria if there are very strong reasons. Appropriate precautions should be taken for all porphyric patients.

Lidocaine solution for injection, in multidose vials containing methyl parahydroxybenzoate (methylparaben) as preservative and should therefore not be used for anaesthesia by the intrathecal, intracisternal or intra- or retro-bulbar routes.

Patients with myasthenia gravis are particularly susceptible to the effects of local anaesthetics.

The effect of local anaesthetics may be reduced if the injection is made into an inflamed or infected area.

Lidocaine injection is not recommended for use in neonates. The optimum serum concentration of lidocaine required to avoid toxicity, such as convulsions and cardiac arrhythmias, in this age group is not known.

INTERACTIONS WITH OTHER MEDICAMENTS

Effects of Lidocaine on other medicinal products

Lidocaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local

anaesthetics (e.g., antiarrhythmics, such as mexiletine), since the systemic toxic effects are additive. There are no literature reports on specific interaction studies with lidocaine and class III antiarrhythmic drugs (e.g., amiodarone), but caution is advised.

There may be an increased risk of enhanced and prolonged neuromuscular blockade in patients treated concurrently with muscle relaxants (e.g., suxamethonium).

Effects of other medicinal products on Lidocaine

There may be an increased risk of ventricular arrhythmia in patients treated concurrently with antipsychotics which prolong or may prolong the QT interval (e.g. pimozide, sertindole, olanzapine, quetiapine, zotepine), or 5HT3 antagonists (e.g. tropisetron, dolasetron).

Concomitant use of quinupristin/dalfopristin should be avoided.

Hypokalaemia produced by acetazolamide, loop diuretics and thiazides antagonises the effect of lidocaine.

The clearance of lidocaine may be reduced by betaadrenoceptor blocking agents (e.g. propranolol) and by cimetidine, requiring a reduction in the dosage of lidocaine.

Increase in serum levels of lidocaine may also occur with anti-viral agents (e.g. amprenavir, atazanavir, darunavir, lopinavir).

Cardiovascular collapse has been reported following the use of bupivacaine in patients on treatment with verapamil and timolol; lidocaine is closely related to bupivacaine.

While adrenaline (epinephrine) when used in conjunction with lidocaine might decrease vascular absorption, it greatly increases the danger of ventricular tachycardia and fibrillation if accidentally injected intravenously.

INCOMPATIBILITIES

Alkalinisation can cause precipitation as lidocaine is limitedly soluble at pH above 6.5. Lidocaine causes precipitation of amphotericin, methohexitone sodium and sulfadiazine sodium in glucose injection. It is recommended that admixtures of lidocaine and glyceryl trinitrate should be avoided.

PREGNANCY AND LACTATION

Pregnancy

Lidocaine passes the placenta. Therefore, lidocaine should not be administered during early pregnancy unless the benefits are considered to outweigh the risks.

Lidocaine given by local perineal infiltration prior to delivery crosses rapidly into the foetal circulation. Elevated lidocaine levels may persist in the newborn for at least 48 hours after delivery. Paracervical blockade or pudendal blockade with lidocaine increases the risk for reactions of the foetus, such as bradycardia/tachycardia, hypotonia or respiratory depression may occur. Therefore, the heart rate of the foetus must be carefully monitored.

Lactation

Small amounts of lidocaine is excreted into breastmilk in small amounts and the possibility of an allergic reaction in the infant, albeit remote, should be borne in mind. The use of lidocaine at recommended doses is unlikely to affect the neonate. Breastfeeding can therefore be continued during use of lidocaine.

ADVERSE REACTIONS

In common with other local anaesthetics, adverse reactions to lidocaine are rare and are usually the result of raised plasma concentrations due to accidental intravascular injection, excessive dosage or rapid absorption from highly vascular areas, or may result from a hypersensitivity, idiosyncrasy or diminished tolerance on the part of the patient. Systemic toxicity mainly involves the central nervous system and/or the cardiovascular system.

Solutions of lidocaine which contain preservatives are not suitable for spinal, epidural or caudal anaesthesia. Adverse effects reported following unpreserved lidocaine solutions administered by this route include hypotension and isolated cases of bradycardia and cardiac arrest.

Immune system disorders

Hypersensitivity reactions (allergic or anaphylactoid reactions, anaphylactic shock). Skin testing for allergy to Lidocaine is not considered to be reliable.

Nervous & Psychiatric disorders

Neurological signs of systemic toxicity include dizziness or light-headedness, nervousness, tremor, circumoral paraesthesia, tongue numbness, drowsiness, convulsions, coma, hyperacusis, visual disturbances, tremor, tinnitus, CNS depression, dysarthria, agitation, euphoria, yawning, restlessness, excitement, muscle twitching followed by depression. Nervous system reactions may be excitatory and/or depressant. Signs of CNS stimulation may be brief, or may not occur at all, so that the first signs of toxicity may be confusion and drowsiness, followed by coma and respiratory failure.

Neurological complications of spinal anaesthesia include transient neurological symptoms such as pain of the lower back, buttock and legs. These symptoms usually develop within twenty-four hours of anaesthesia and resolve within a few days. Isolated cases of arachnoiditis or cauda equina syndrome, with persistent paraesthesia, bowel and urinary dysfunction, or lower limb paralysis have been reported following spinal anaesthesia with lidocaine and other similar agents. The majority of cases have been associated with hyperbaric concentrations of lidocaine or prolonged spinal infusion. Neuropathy, peripheral nerve injury.

Eye disorders

Blurred vision, diplopia and transient amaurosis may be signs of lidocaine toxicity. Bilateral amaurosis may also be a consequence of accidental injection of the optic nerve sheath during ocular procedures. Orbital inflammation and diplopia have been reported following retro- or peribulbar anaesthesia. Double vision.

Ear and labyrinth disorders

Tinnitus, hyperacusis.

Cardiac and vascular disorders

Hypertension. Cardiovascular reactions are depressant and may manifest as hypotension, bradycardia, myocardial depression, cardiac arrhythmias and possibly cardiac arrest or circulatory collapse.

Respiratory, thoracic or mediastinal disorders

Dyspnoea, bronchospasm, respiratory depression, respiratory arrest.

Gastrointestinal disorders

Nausea, vomiting.

Skin & subcutaneous tissue disorders

Rash, urticaria, oedema (including angioedema, face oedema).

Blood and the lymphatic system disorders

Methaemoglobinaemia.

General Disorders and Administration Site Conditions

Infusion site swelling.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Accidental intravascular injections of local anaesthetics can cause immediate system-toxic reactions (within seconds to a few minutes). Signs of system toxicity in cases of overdose occur later (15-60 minutes after injection) due to a slower increase in the concentration of local anaesthetic in the blood.

Toxicity:

Parenteral administration: 50 mg intravenously to a 1 months old child caused very severe intoxication. 200-400 mg infiltration to an adult caused serious, 500 mg to an 80 year old patient and 1 g intravenously to adults caused very severe intoxication.

Symptoms:

First CNS excitation, later CNS depression. When exposed for large doses the first symptom can be rapidly developing seizures. Agitation, dizziness, visual disturbances, perioral paresthesia, nausea, numbness of the tongue, hyperacusis, and tinnitus. Later ataxia, auditory disturbances, euphoria, confusion, speech difficulties, paleness, sweating, tremor, seizures, coma, apnoea. Visual disturbance and muscular tremors or muscle twitching are more serious and precede the onset of generalised convulsions. These signs must not be mistaken for neurotic behaviour. Unconsciousness and grand mal convulsions may follow, which may last from a few seconds to several minutes. Hypoxia and hypercapnia occur rapidly following convulsions due to increased muscular activity, together with the interference with normal respiration and loss of the airway. Acidosis increases the toxic effects of local anaesthetics. Different kinds of arrhythmias especially bradycardial arrhythmias but also if exposed to

large doses ventricular tachycardia, ventricular fibrillation, QRS prolongation, atrio-ventricular block, cardiac failure, hypotension, with potentially fatal outcome. (rare cases of methaemoglobinaemia are described). Recovery occurs as a consequence of redistribution of the local anaesthetic drug from the central nervous system, and metabolism and may be rapid unless large amounts of the drug have been injected.

Treatment:

If signs of acute systemic toxicity appear, injection of the anaesthetic should be stopped immediately. Treatment will be required if convulsions and CNS depression occurs. The objectives of treatment are to maintain oxygenation, stop the convulsions and support the circulation. If ventricular lavage is indicated, this should be done with a ventricular tube after tracheal intubation). Seizures are treated with diazepam. Prolonged convulsions may jeopardise the patient's ventilation and oxygenation and early endotracheal intubation should be considered. Oxygen. If needed tracheal intubation and controlled respiration (if needed hyperventilation). Bradycardia is treated with atropine. Circulatory deterioration is treated with fluids administered intravenously, dobutamine and if needed epinephrine (initially 0.05 microg/kg BW/min, to be increased if needed by 0.05 microg/kg BW/min each 10 minutes.), in more severe cases guided by haemodynamic monitoring. Ephedrine may also be tried. If circulatory arrest, resuscitation efforts for several hours may be indicated. Where further supportive treatment of circulatory depression is required, use of a vasopressor agent may be considered although this involves a risk of central nervous system excitation. Continual optimal oxygenation and ventilation and circulatory support as well as treatment of acidosis are of vital importance. Dialysis is of negligible value in the treatment of acute overdosage with lidocaine.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Depending on the dose and method of administration, lidocaine can have a transient effect on movement and coordination.

Where outpatient anaesthesia affects areas of the body involved in driving or operating machinery, patients should be advised to avoid these activities until normal function is fully restored.

INSTRUCTION FOR USE

Restertilisation of lidocaine is not recommended.

Appropriate control procedures to prevent contamination should be employed, including the following:

- use of single-use sterile injecting equipment;
- use a sterile needle and syringe for each new insertion into the vial; rule out the introduction of contaminated material or fluid into a multidose vial.

PACKING / PACK SIZE

10 x 10mL vial (clear)

STORAGE CONDITIONS

Store below 30°C. Protect from light. Retain in carton until time of use. Do not use if solution contain particles, growth, turbidity or if there is any change of appearance.

Once opened, this product is stable for 72 hours at both refrigerated condition (0°C – 10°C) and under 30°C storage condition.

SHELF LIFE

3 years from date of manufacture (unopened).

Once opened, this product is stable for 72 hours at both refrigerated condition (0°C – 10°C) and under 30°C storage condition.

**PRODUCT REGISTRATION HOLDER /MANUFACTURER:
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